Title

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Abstract

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1 The setup

Recall that we have a line bundle $\mathcal{L} := \mathcal{L}_{\det}$ on $\mathcal{X} := \mathcal{X}_{G,g,I}$ and an invariant norm $\|\cdot\|$ on $X_*(T)_{\mathbb{R}}$.

For a point $x \in \mathcal{X}(k)$ and a nontrivial $f : \Theta \to \mathcal{X}$ with $f(1) \simeq x$, define

$$\mu(x, f) \ := \ \frac{\mathrm{wt}_{\mathcal{L}}(f)}{\|\lambda_f\|},$$

where $\mathrm{wt}_{\mathcal{L}}(f)$ is the $\mathbb{G}_m$-weight on the fiber of $\mathcal{L}$ at the special point $f(0)$, and λ_f is the associated cocharacter data of f coming from the action of $\mathbb{G}_m$ on the object $f(0)$.

We wanted to prove the following theorem.

Theorem 1.1 (Need to prove)**.** For any $x \in \mathcal{X}_{G,g,I}(k)$, there exists a (unique up to something) maximally destabilizing Θ-filtration $f_{\mathrm{HN}} : \Theta \to \mathcal{X}_{G,g,I}$ of x with respect to $\mathcal{L}_{\det}$ and any invariant norm on $X_*(T)_{\mathbb{R}}$.

Constantin remarked that the uniqueness part of the theorem is not important for our application, and that we can get away with existence of a maximally destabilizing Θ-filtration for each $x \in \mathcal{X}(k)$, provided that we can organize the points x into strata in such a way which mirrors the setup of the Kirwan Ness stratification.

2 Kirwan Ness stratification

This section follows [**?**]. Suppose we have linear action of a reductive group G on a projective variety X, singular or nonsingular, defined over an algebraically closed field. Let $T \subset G$ be a maximal torus, V the representation giving the linearization. Pick an invariant norm on the cocharacter lattice of G. Then Kirwan and Ness construct a stratification of X into locally closed subvarieties

$$X = \bigsqcup_{\beta \in \mathcal{B}} S_\beta,$$

The index set $\mathcal{B}$ is in correspondence with connected components $Z \subset X^\beta$ of the fixed locus of a dominant cocharacter β of T such that the semistable locus Z° of the divided action of the Levi subgroup L_β on Z is nonempty.

If the fixed loci is connected, then the index set $\mathcal{B}$ can be identified with the set of G conjugacy classes of rays in $X_*(G) \otimes \mathbb{R}$ together with the condition that the associated "center" Z_β^{ss} is nonempty. Note that every cocharacter of G is G-conjugate to a dominant cocharacter of T. It is harmless to choose a rational dominant cocharacter β in each conjugacy class as they determine the same parabolic subgroup P_β and the same stratum S_β. Note that one cannot simply group the components together under a single conjugacy class of rays, as each connected component Z_i could possibly a different weight of the linearized line bundle.

Then we have a weight decomposition

$$V = \bigoplus_{\chi} V_\chi,$$

and so for any point $x \in X$, we can write its homogeneous coordinates as $x = [v]$ for some $v \in V$ with $v = \sum_\chi v_\chi$. For a point x, let W_x be the set of weights appearing in its support, meaning the set of χ such that $v_\chi \neq 0$. Then Kirwan identifies β as the closest point to 0 in $\mathrm{Conv}(W_x)$.

Alternatively, we can identify β as the G-conjugacy class of ray which minimzes the following

function on $X_*(G) \otimes \mathbb{R}$:

$$\lambda \mapsto \frac{\mu(x, \lambda)}{\|\lambda\|}.$$

where $\mu(x, \lambda)$ is defined as the minimum λ-weight of the nonzero coordinates of x.

$$\mu(x, \lambda) = \min_{\chi \in W_x} \langle \chi, \lambda \rangle$$

Proposition 2.1 (Hilbert-Mumford criterion for semistability)**.** A point x is semistable if and only if $\mu(x, \beta) \geq 0$ for all dominant cocharacters β.

For each $\beta \in \mathcal{B}$, pick a dominant rational representative of the corresponding conjugacy class of rays which we also denote by β. The unstable strata are indexed by those Z with dominant β for which the semistable locus $Z^\circ \subset Z$ of the divided L-action on $\mathcal{L}$ is not empty.

In particular we look at the fixed locus $X^\beta = \{x \in X \mid \beta(t) \cdot x = x \ \forall t\}$ and take a connected component $Z \subset X^\beta$. Now define the attracting set:

$$Y = \{x \in X \mid \lim_{t \to \infty} \lambda(t) \cdot x \in Z\}$$

which gives us a natural map $\varphi : Y \to Z$ defined by the limit of the T-flow.

Proposition 2.2. $\varphi : Y \to Z$ is a locally trivial fibration in affine spaces. At a point $z \in Z$, the tangent space decomposes into weight spaces under β as:

$$T_z X = T_z Z \oplus (T_z X)_{>0}$$

The positive-weight directions integrate to affine fibers.

Now we describe what Z° is. Let $L = Z_G(\beta)$ be the Levi subgroup. β acts on the fiber of the linearized line bundle $\mathcal{L}$ over Z by the Hilbert-Mumford weight $\mu(x, \beta)$. Let χ be the character of L given by this action. Twisting down by χ, the action of L on Z is trivial on the fiber of the linearized line bundle, and so we can talk about semistability for the L-action on Z with respect to the trivial linearization. We define Z° to be the semistable locus for this GIT problem, and put $Y^\circ = \varphi^{-1}(Z^\circ)$. Note that if we did not twist down by χ, then the action on the fiber would be nontrivial and so the semistable locus would be empty, since $\beta : \mathbb{G}_m \to L$ is a one-parameter subgroup of L which violates the Hilbert-Mumford criterion for semistability.

To recover the stratum S_β, we take the G-orbit of Y°, i.e. $S = G \cdot Y^\circ$ and then one can show that we have the following isomorphism of G-varieties:

$$S \cong G \times^{P_\beta} Y^\circ$$

where we are dividing by the relation

$$(g, y) \sim (gp^{-1}, py) \quad \forall p \in P_\beta.$$

Proposition 2.3. In this setup, we have the following properties:

(i) [Bialynicki-Birula decomposition] Y is a fiber bundle over Z, with affine spaces as fibers, under the morphism φ defined by the limiting value of the T-flow. The fibers of the affine bundle are the positive weight spaces in the tangent space of X at the limit point.

(ii) Y is stabilized by the parabolic subgroup $P \subset G$ whose nilpotent Lie algebra radical $\mathfrak{u}$ is spanned by the negative T-eigenspaces in $\mathfrak{g}$.

(iii) S is equivariantly isomorphic to $G \times^P Y^\circ$. In particular

$$S \cong G \times^P Y^\circ \cong (G \times^P Z^\circ) \times \mathbb{A}^n$$

where the fibers of the affine bundle are the positive weight spaces in the tangent space of X at the limit point, if we let P act on Z° via its reductive quotient L.

(iv) The various S, together with $X^\circ = X^{ss}$, smoothly stratify X.

(v) Z° has a projective, good quotient under L; X^{ss} has a good projective quotient under G.

3 Lattices in $\mathrm{SL}(2)$

Let $F = \mathbb{C}((z))$, $\mathcal{O} = \mathbb{C}[[z]]$, $G = \mathrm{SL}_2(F)$. Let

$$\Lambda_0 := \mathcal{O}e_1 \oplus \mathcal{O}e_2, \qquad \Lambda_1 := \mathcal{O}e_1 \oplus z\,\mathcal{O}e_2,$$

and

$$P_0 := \mathrm{Stab}_G(\Lambda_0) = \mathrm{SL}_2(\mathcal{O}), \qquad P_1 := \mathrm{Stab}_G(\Lambda_1).$$

Let

$$S := \begin{pmatrix} 1 & 0 \\ 0 & z \end{pmatrix} \in \mathrm{GL}_2(F).$$

Then $S\Lambda_0 = \Lambda_1$. Hence

$$P_1 = \mathrm{Stab}_G(\Lambda_1) = \mathrm{Stab}_G(S\Lambda_0) = S\ \mathrm{Stab}_G(\Lambda_0)\, S^{-1} = SP_0S^{-1},$$

because conjugation by S preserves determinant 1, so $SP_0S^{-1} \subset \mathrm{SL}_2(F)$ and it is exactly the stabilizer of $S\Lambda_0$.

Conjugation by S gives a bijection of double coset sets

$$\begin{aligned} P_1\backslash G/P_1 &\xrightarrow{\sim} P_0\backslash G/P_0, \\ P_1gP_1 &\mapsto P_0(S^{-1}gS)P_0. \end{aligned}$$

Moreover we know that $P_0\backslash G/P_0$ is in bijection with the set of dominant cocharacters of SL_2, which is $\mathbb{Z}_{\geq 0}$, via the Cartan decomposition

$$G = \bigsqcup_{n\geq 0} P_0 \begin{pmatrix} z^n & 0 \\ 0 & z^{-n} \end{pmatrix} P_0.$$

Hence $P_1\backslash G/P_1$ is also in bijection with $\mathbb{Z}_{\geq 0}$.

3.1 Heinloth's weight formula for SL_2 and the nodal contribution on X_{Solis}

Let C be a smooth projective curve and E a principal SL_2–bundle on C. Let $\mathcal{L}_{\det}$ denote the determinant of cohomology line bundle on $\mathrm{Bun}_{\mathrm{SL}_2}(C)$ associated to the adjoint representation Ad.

1. Smooth case: Heinloth's formula for SL_2

A maximal parabolic of SL_2 is a Borel subgroup $B \subset \mathrm{SL}_2$, so a reduction E_B is equivalently a line subbundle

$$L \subset V := E \times^{\mathrm{SL}_2} \mathbb{C}^2 \qquad \text{with} \qquad \det(V) \cong \mathcal{O}_C.$$

Let $T \subset B$ be a maximal torus. As a T–representation,

$$\mathfrak{sl}_2 = \mathfrak{t} \oplus \mathfrak{g}_+ \oplus \mathfrak{g}_-,$$

where $\mathfrak{g}_+$ (resp. $\mathfrak{g}_-$) is the positive (resp. negative) root space and $\mathfrak{t}$ has weight 0. Under the Borel reduction determined by L, the positive root bundle is

$$E_B \times^B \mathfrak{g}_+ \cong L^{\otimes 2}.$$

Let $f : \Theta = [\mathbb{A}^1/\mathfrak{G}_m] \to \mathrm{Bun}_{\mathrm{SL}_2}(C)$ be the very close degeneration $Rees(E_B, \check{\omega})$ attached to the fundamental cocharacter $\check{\omega}$ (the unique maximal parabolic direction for SL_2). Heinloth's computation ([**?**, §1.F.c]) gives

$$\mathrm{wt}_{\mathcal{L}_{\det}}(f) = 2\sum_{i>0} i \cdot \deg\big(E_{0,\check{\omega}} \times^{L_{\check{\omega}}} \mathrm{Lie}(G)_i\big) = 2\deg(L^{\otimes 2}) = 4\deg(L), \tag{1}$$

up to the choice of convention $t \to 0$ versus $t \to \infty$ in the definition of the weight.

2. The nodal/bubble contribution on X_{Solis}

Now let $x \in D_0 \subset X_{\mathrm{Solis}}$ be a hyperspecial boundary point. Thus x is represented by an admissible SL_2–bundle E' on a twisted modification

$$m : C' \longrightarrow C_0$$

of the nodal curve C_0, obtained by inserting an exceptional chain over the node. Let $f : \Theta \to X_{\mathrm{Solis}}$ be a very close degeneration with $f(1) = x$ and special fiber $f(0)$ an associated graded object carrying a Borel reduction E'_B. Let

$$L' \subset V' := E' \times^{\mathrm{SL}_2} \mathbb{C}^2$$

be the corresponding line subbundle on C' (so $\det(V') \cong \mathcal{O}_{C'}$). The same representation–theoretic identification as above gives

$$E'_B \times^B \mathfrak{g}_+ \cong (L')^{\otimes 2}.$$

Therefore Heinloth's degree formula applies verbatim, but with degree computed on C':

$$\mathrm{wt}_{\mathcal{L}_{\det}}(f) = 2 \deg_{C'}\big((L')^{\otimes 2}\big) = 4 \deg_{C'}(L'). \tag{2}$$

Write C' as the union of the strict transform of the normalization and the exceptional chain:

$$C' = \widetilde{C} \cup (\text{exceptional chain}),$$

with intersection along the two attaching points. Additivity of degree gives

$$\deg_{C'}(L') = \deg_{\widetilde{C}}\big(L'|_{\widetilde{C}}\big) + \deg_{\mathrm{chain}}\big(L'|_{\mathrm{chain}}\big). \tag{3}$$

The first term is the usual Harder–Narasimhan/Rees contribution coming from the smooth part. The second term is the nodal/bubble contribution.

In the hyperspecial stratum D_0, the local gluing class at the node is labeled by a dominant coweight

$$\lambda \in X_*(T)_+ \cong \mathbb{Z}_{\geq 0}.$$

Geometrically, the associated graded local model is a bubble carrying the split bundle

$$\mathcal{O}_{\mathbb{P}^1}(\lambda) \oplus \mathcal{O}_{\mathbb{P}^1}(-\lambda),$$

and the graded Borel reduction picks out the summand $\mathcal{O}_{\mathbb{P}^1}(\lambda)$. Hence on each bubble component we have

$$L'|_{\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}(\lambda), \qquad \text{so} \qquad \deg_{\mathbb{P}^1}(L'|_{\mathbb{P}^1}) = \lambda.$$

For an exceptional chain, the same statement holds with λ equal to the *total* relative position across the chain (distributed among components), so

$$\deg_{\mathrm{chain}}\big(L'|_{\mathrm{chain}}\big) = \lambda. \tag{4}$$

Combining (2), (3), and (4) yields the advertised decomposition of the weight:

$$\mathrm{wt}_{\mathcal{L}_{\det}}(f) = 4\Big(\deg_{\widetilde{C}}\big(L'|_{\widetilde{C}}\big) + \lambda\Big), \qquad (\mathrm{SL}_2,\ x \in D_0). \tag{5}$$

The Θ-degeneration determines a B-reduction E'_B of the graded object E' on C', L' is the line bundle associated to that B-reduction (equivalently the line subbundle $L' \subset V'$), and the nodal contribution is $\deg(L'|_{\mathrm{chain}})$, which can vary with the reduction even when the ambient hyperspecial double-coset label λ is fixed.

A. Degree on a nodal curve: normalization + a len correction

Let C_0 be an integral projective nodal curve with a single node $p \in C_0$, and let $\nu : \widetilde{C} \to C_0$ be the normalization. Write $\nu^{-1}(p) = \{p_x, p_y\}$.

Lemma 3.1 (Degree of a rank-1 torsion-free sheaf at a node)**.** Let F be a rank-1 torsion-free coherent sheaf on C_0. Then $\widetilde{F} := \nu^* F$ is a line bundle on $\widetilde{C}$, and there is a canonical exact sequence

$$0 \longrightarrow F \longrightarrow \nu_* \widetilde{F} \longrightarrow Q \longrightarrow 0, \tag{6}$$

where Q is a skyscraper sheaf supported at p. Moreover,

$$\deg_{C_0}(F) \;=\; \deg_{\widetilde{C}}(\widetilde{F}) \;+\; \mathrm{len}(Q). \tag{7}$$

Proof. Since F is rank-1 torsion-free on a nodal curve, it is locally free away from p, and pulling back to the normalization removes the singularity, so $\widetilde{F} = \nu^* F$ is a line bundle on $\widetilde{C}$. The natural adjunction map $F \to \nu_* \nu^* F = \nu_* \widetilde{F}$ is an isomorphism away from p, hence its cokernel Q is supported at p, giving (6).

Taking Euler characteristics in (6) yields

$$\chi(C_0, F) = \chi(C_0, \nu_* \widetilde{F}) - \chi(C_0, Q) = \chi(\widetilde{C}, \widetilde{F}) - \mathrm{len}(Q),$$

since $\chi(C_0, \nu_* \widetilde{F}) = \chi(\widetilde{C}, \widetilde{F})$ and $\chi(C_0, Q) = \mathrm{len}(Q)$. The same argument with $F = \mathcal{O}_{C_0}$ gives

$$\chi(C_0, \mathcal{O}_{C_0}) = \chi(\widetilde{C}, \mathcal{O}_{\widetilde{C}}) - \mathrm{len}(\nu_* \mathcal{O}_{\widetilde{C}} / \mathcal{O}_{C_0}),$$

and $\mathrm{len}(\nu_* \mathcal{O}_{\widetilde{C}} / \mathcal{O}_{C_0}) = 1$ for a node. Using the definition $\deg_X(G) := \chi(X, G) - \chi(X, \mathcal{O}_X)$ for rank-1 torsion-free G on a proper curve X, the len terms combine to give (7). (Equivalently, one may take (7) as the definition of $\deg_{C_0}(F)$ for torsion-free rank-1 sheaves.) $\square$

B. Interpreting the hyperspecial invariant λ in the D_0 case

In the hyperspecial stratum D_0 we do *not* necessarily insert an exceptional chain of $\mathbb{P}^1$'s. Instead the local invariant

$$\lambda \in X_*(T)_+ \cong \mathbb{Z}_{\geq 0}$$

is read off from the hyperspecial double coset $\mathrm{SL}_2(\mathcal{O}) \backslash \mathrm{SL}_2(F) / \mathrm{SL}_2(\mathcal{O})$ via the Cartan decomposition:

$$g \in \mathrm{SL}_2(\mathcal{O}) \begin{pmatrix} z^{\lambda} & 0 \\ 0 & z^{-\lambda} \end{pmatrix} \mathrm{SL}_2(\mathcal{O}).$$

Let $f : \Theta \to X_{\mathrm{Solis}}$ be a very close degeneration with $f(1) = x$ and let $E'^{\mathrm{gr}} = f(0)$ be the associated graded object. The Θ-limit supplies a Borel reduction E'^{gr}_B, hence a line subsheaf

$$L \subset V := E'^{\mathrm{gr}} \times^{\mathrm{SL}_2} \mathbb{C}^2$$

of the associated rank-2 bundle on C_0. In general L is only rank-1 torsion-free at the node. Applying Lemma 3.1 to $F = L$ gives

$$\deg_{C_0}(L) \;=\; \deg_{\widetilde{C}}(\nu^* L) \;+\; \mathrm{len}(Q_L), \qquad Q_L := \nu_* \nu^* L / L. \tag{8}$$

Remark 3.2 (Where λ enters)**.** For the *associated graded* local model of type λ (i.e. after conjugating the gluing to the diagonal representative and taking the Borel line picked out by the Θ-limit), the failure of L to be locally free at the node is measured precisely by λ, in the sense that

$$\mathrm{len}(Q_L) = \lambda.$$

Thus in the D_0 case (with no exceptional chain inserted) the "nodal term" appearing in the weight computation is a *len correction at the node*, not a degree on a bubble.

C. Consequence for the $\mathcal{L}_{\det}$-weight in the D_0 case

Heinloth's formula gives, for SL_2 and the adjoint determinant line bundle,

$$\mathrm{wt}_{\mathcal{L}_{\det}}(f) = 4\,\deg_{C_0}(L)$$

(up to the $t \to 0$ versus $t \to \infty$ convention). Combining with (8) and Remark 3.2 yields

$$\mathrm{wt}_{\mathcal{L}_{\det}}(f) \;=\; 4\Big(\deg_{\widetilde{C}}(\nu^* L) \;+\; \lambda\Big), \qquad (\mathrm{SL}_2,\ x \in D_0).$$

3.2 The D_1 case

Changing the trivializations on each branch by parahoric gauge transformations $p_x \in \mathcal{P}_x$, $p_y \in \mathcal{P}_y$ changes γ by

$$\gamma \mapsto p_y^{-1}\,\gamma\,p_x.$$

Let E_x be a $\mathcal{P}_x$-torsor on the x-disc and E_y a $\mathcal{P}_y$-torsor on the y-disc.

- Over the punctured discs they become $G((z))$-torsors (since $\mathcal{P}_\bullet \hookrightarrow G((z))$).
- Choose trivializations on the punctured discs:

$$\tau_x : E_x|_{D_x^\times} \xrightarrow{\sim} G((z)),$$
$$\tau_y : E_y|_{D_y^\times} \xrightarrow{\sim} G((z)).$$

- The gluing isomorphism (descent datum) is an isomorphism of $G((z))$-torsors

$$\phi : E_x|_{D_x^\times} \xrightarrow{\sim} E_y|_{D_y^\times}.$$

Let us explain how to model descent datum on the nodal curve. Let $\mathcal{P} = \mathcal{L} \ltimes \mathcal{U}$ be the parahoric with Levi $\mathcal{L}$ and pro-unipotent radical $\mathcal{U}$. Solis sets

$$P^{\Delta} := \Delta(\mathcal{L}) \ltimes (\mathcal{U} \times \mathcal{U}) \subset LG \times LG$$

Consider the map

$$\Phi : LG \times LG \longrightarrow LG, \qquad (g_1, g_2) \longmapsto g_2\, g_1^{-1}.$$

If (g_1, g_2) is changed by an element of P^{Δ}, say

$$(g_1, g_2) \mapsto (g_1 u_1 \ell,\ g_2 u_2 \ell) \quad \text{with } u_i \in U,\ \ell \in L,$$

then

$$g_2 u_2 \ell \cdot (g_1 u_1 \ell)^{-1} = g_2\, (u_2\, \ell\, \ell^{-1}\, u_1^{-1})\, g_1^{-1} = g_2\, u_2\, u_1^{-1}\, g_1^{-1}.$$

So the Levi cancels and Φ changes by

$$\Phi \mapsto u_2 \cdot \Phi \cdot u_1^{-1}.$$

Conclusion: Φ descends to a map of quotients

$$(LG \times LG)/\mathcal{P}^{\Delta} \longrightarrow U\backslash LG/U.$$

This quotient is the framed version of the local moduli at the node:

1. g_1 is a trivialization on the u-branch punctured disc,
2. g_2 is a trivialization on the v-branch punctured disc,
3. modding out by $\mathcal{U} \times \mathcal{U}$ is change trivializations by parahoric-unipotent gauge on each branch.
4. modding out by $\Delta(\mathcal{L})$ forces the Levi parts on the two branches to match (this is the descent compatibility across the node).

We can further quotient

$$U\backslash LG/U \twoheadrightarrow \mathcal{P}\backslash LG/\mathcal{P}$$

because $\mathcal{P} = L \ltimes U$ and this recovers the relative position invariants $\mathcal{P}\backslash LG/\mathcal{P} \cong W_I\backslash\widetilde{W}/W_I$ from which we will extract the nodal contribution to the weight.

The local parameter space before taking invariants is refined

$$(LG \times LG)/P^{\Delta} \to U\backslash LG/U \to \mathcal{P}\backslash LG/\mathcal{P}$$

The local node chart $(LG \times LG)/\mathcal{P}^{\Delta}$ refines the relative-position in $\mathcal{P}\backslash LG/\mathcal{P} \simeq W_I\backslash\widetilde{W}/W_I$. A Θ-filtration supplies a 1-PS $\lambda : \mathbb{G}_m \to \Delta(L)$ which acts with strictly positive weights on the pro-unipotent factor U. Therefore the $\mathbb{G}_m$-limit $t \to 0$ contracts the U-coordinates in the corresponding

Bruhat/Schubert cell, and the limit lands in the unique T-fixed point in the Schubert closure, i.e. the point indexed by the minimal-len double-coset representative $w_{\min} \in W_I \backslash \widetilde{W} / W_I$.

This aids our calculation of the Hilbert Mumford weight as follows. Given a Θ-degeneration $f : \Theta \to X_{\text{Solis}}$. Over $\mathbb{A}^1$ you get a family of line bundles (your L' coming from the reduction, or the positive-root line $(L')^{\otimes 2}$ in ad). The degree on the exceptional chain is constant in flat families, so you may compute it on the special fiber $t = 0$.

So you are allowed to replace your original local point (with complicated U-coordinates) by its $\mathfrak{G}_m$-limit (the associated graded) without changing the degree you are computing.

Locally at the node, your object gives a point in a Schubert stratum of the partial affine flag variety $\mathrm{Fl}_I = LG/\mathcal{P}$ (or a refinement thereof). Flowing by the cocharacter λ contracts U and lands at some T-fixed point in the Schubert closure. The T-fixed points in $LG/\mathcal{P}$, under the action of T on $LG/\mathcal{P}$ via left multiplication, are precisely $\{w\mathcal{P}\}_{w \in \widetilde{W}}$, where $w\mathcal{P} \in LG/\mathcal{P}$.

I suspect that the point indexed by the minimal double-coset representative $w_{\min} \in W_I \backslash \widetilde{W} / W_I$.

Thus the local geometry of your degeneration becomes "the T-fixed point $w_{\min}$" plus a chain of $\mathbb{P}^1$'s corresponding to the simple reflections appearing in a reduced word for $w_{\min}$.

On Fl_I, the Picard group is generated by the standard line bundles $\mathcal{L}_\alpha$ whose degrees are normalized by

$$\deg(\mathcal{L}_\alpha|_{\mathbb{P}^1_\alpha}) = 1$$

on the basic Schubert curve $\mathbb{P}^1_\alpha$ corresponding to a simple affine reflection $\alpha \notin I$.

So if your local line bundle (coming from the fundamental weight/Borel reduction) is the pullback of one of these generators (or a known combination), then its degree on any 1–cycle is computed by counting how many times that 1–cycle traverses the corresponding $\mathbb{P}^1_\alpha$'s. Take a reduced expression $w_{\min} = s_{\alpha_1} s_{\alpha_2} \cdots s_{\alpha_r}$ in simple affine reflections (with $\alpha_i \notin I$ in the partial flag setting). The corresponding Schubert variety contains a canonical chain of Schubert curves

$$\mathbb{P}^1_{\alpha_1} \cup \mathbb{P}^1_{\alpha_2} \cup \cdots \cup \mathbb{P}^1_{\alpha_r}$$

connecting the base point to $w_{\min}$.

Then for a line bundle $\mathcal{L}$ whose degree on $\mathbb{P}^1_\alpha$ is known, you get

$$\deg(\mathcal{L} \text{ on that chain}) = \sum_{j=1}^{r} \deg\big(\mathcal{L}|_{\mathbb{P}^1_{\alpha_j}}\big).$$

In SL_2, D_1 case there is only one surviving simple direction (say α_0), so every reduced word for $w_{\min}$ uses α_0 exactly n times, and

$$\deg(\mathcal{L}|_{\text{chain}}) = n \cdot \deg(\mathcal{L}|_{\mathbb{P}^1_{\alpha_0}}).$$

With the standard normalization $\deg(\mathcal{L}|_{\mathbb{P}^1_{\alpha_0}}) = 1$, this is just n.

That is exactly why the "combinatorial number" attached to $w_{\min}$ becomes the degree.

How this feeds your $\mathcal{L}_{\det}$-weight calculation

Heinloth's weight for SL_2 is

$$\mathrm{wt}_{\mathcal{L}_{\det}}(f) = 2\deg\left(E'_B \times^B \mathfrak{g}_+\right) = 2\deg\left((L')^{\otimes 2}\right) = 4\deg(L').$$

So once you identify $\deg(L'|_{\text{chain}})$ with the integer n coming from $w_{\min}$, the nodal contribution to the weight is immediately $4n$ (up to your sign convention).

If you want the next step prove $\deg(L'|_{\text{chain}}) = n$ the clean proof route is: - identify the chain with the Schubert $\mathbb{P}^1$-chain in the partial affine flag variety attached to $w_{\min}$, - use that $\mathrm{Pic}(LG/P_1) \cong \mathbb{Z}$ and degrees are normalized on the basic Schubert $\mathbb{P}^1$, - and observe that $w_{\min} = (s_0 s_1)^n$ traverses that $\mathbb{P}^1$ exactly n times.

On the node, after taking the associated graded, the local data gives a map from the exceptional chain (a chain of $\mathbb{P}^1$'s) into the relevant Schubert closure in the partial affine flag LG/P_I. Each irreducible component of the chain maps to a Schubert curve $\mathbb{P}^1_\alpha$

Recall we are trying to compute the degree of the determinant of cohomology line bundle associated to the adjoint representation.

4 References

Teleman Quantization Conjecture Revisited. Kirwan Cohomology of quotients in symplectic and algebraic geometry, 1984.